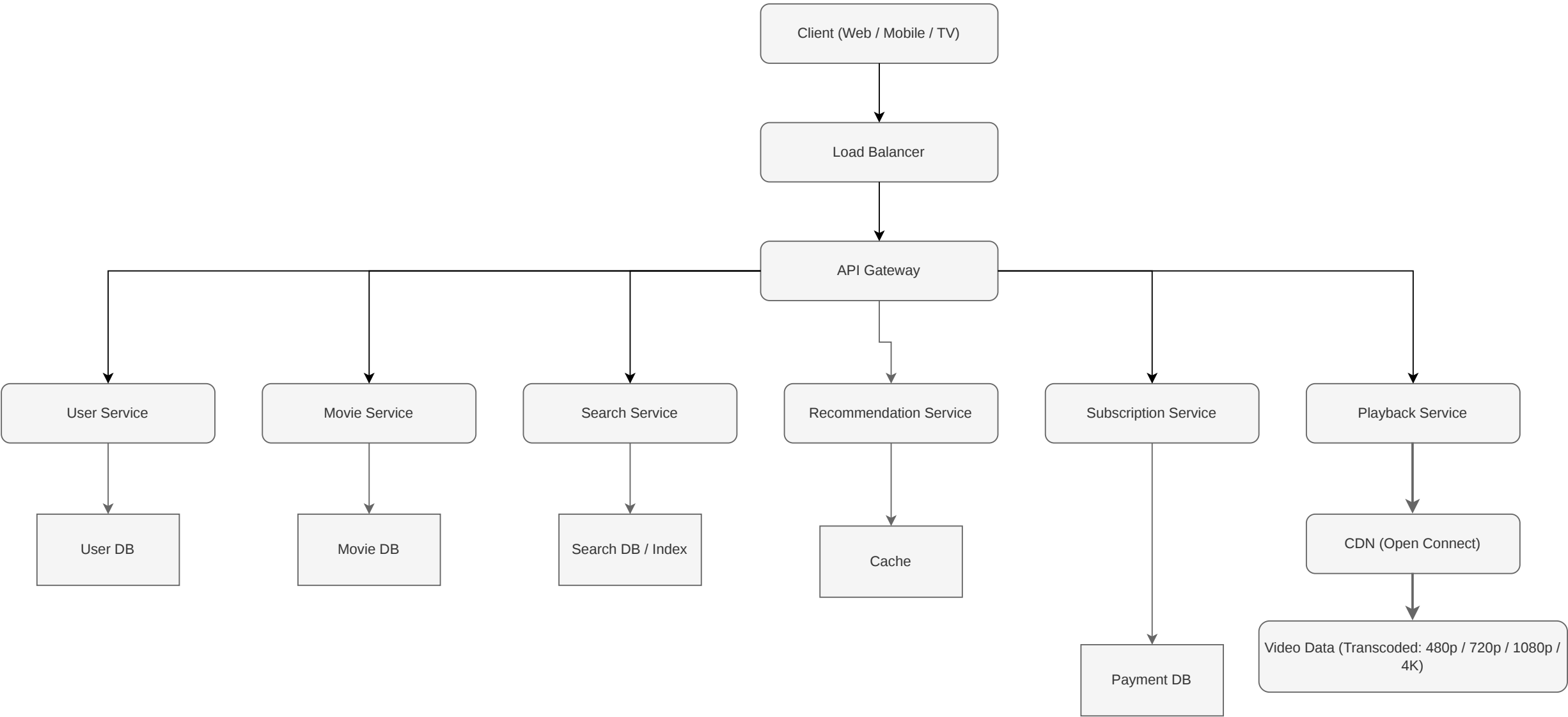




Requirements

Functional Requirements	Non-Functional Requirements
1. Users can register, log in, and manage their profile	1. High availability (target ~99.99% uptime)
2. Users can browse and search the movie/show catalog	2. Low latency video start (< 2 seconds) and playback
3. Users can play and stream video content	3. Scalability to support millions of concurrent streams
4. System checks subscription/payment status before playback	4. Fault tolerance and redundancy (no single point of failure)
5. System provides personalized recommendations	5. Strong consistency for payments; eventual consistency OK elsewhere
6. System tracks watch history and supports Continue Watching	6. Security: encrypted data, auth tokens, DRM for video content
7. Users can rate/review content	7. Adaptive bitrate streaming to handle varying network conditions
8. Playback works across Web, Mobile, and Smart TV	8. Global low-latency delivery via CDN edge locations

Capacity Estimation

Traffic & Users • Total registered users: ~250M • Daily Active Users (DAU): ~100M • Peak concurrent streams: ~15M (assume 15% of DAU streaming at peak) • Avg. sessions/user/day: ~1.5 • Search QPS: ~50K requests/sec at peak • Metadata (browse/home) QPS: ~200K requests/sec at peak	Bandwidth Estimation • Avg. bitrate per stream (HD): ~5 Mbps • Peak bandwidth = 15M streams × 5 Mbps ≈ 75 Tbps (offloaded mostly to CDN edge nodes) • Origin-to-CDN fill bandwidth: much smaller (cache hit ratio > 95%) • Adaptive bitrate ladder: 480p / 720p / 1080p / 4K reduces avg load	Storage Estimation • Catalog size: ~15,000 titles • Avg. size per title (all resolutions, ~2 hrs): ~30 GB • Raw video storage = 15,000 × 30 GB ≈ 450 TB (replicated across regions/CDN → multiple PB total) • Metadata DB: a few hundred GB (titles, cast, genres) • Watch history: ~100M users × ~2 KB/record/day ≈ 200 GB/day
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Caching & Misc

- Recommendation cache: refreshed periodically, sized per-user summary (~few KB × 100M DAU ≈ hundreds of GB)
- Popular-title cache at CDN edge covers majority of watch traffic (long-tail served from origin/regional storage)
- API Gateway: rate limiting + auth checks add small, constant-time overhead per request